

How much does an RL scheduler lose when the traffic changes? Quantifying transfer gaps in RAN slicing

A policy trained on one traffic mix, deployed on another: how big is the penalty, and can we predict it?

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5–8 pts

of SLA satisfaction (SSR) lost, on average, when a policy meets the wrong traffic mix. Worst pair: 21 pts.

≈ 1 : 1

transfer gap vs. seed-to-seed noise (ratio 0.72–1.49). Retraining per scenario is not yet justified.

Balanced wins

Policies trained on balanced traffic transfer best (rank 1 / 2 / 1) and are the only ones with a negative mean gap.

No predictor

Distance between allocation distributions does not predict the gap: correlation flips sign across runs.

1 Problem & approach

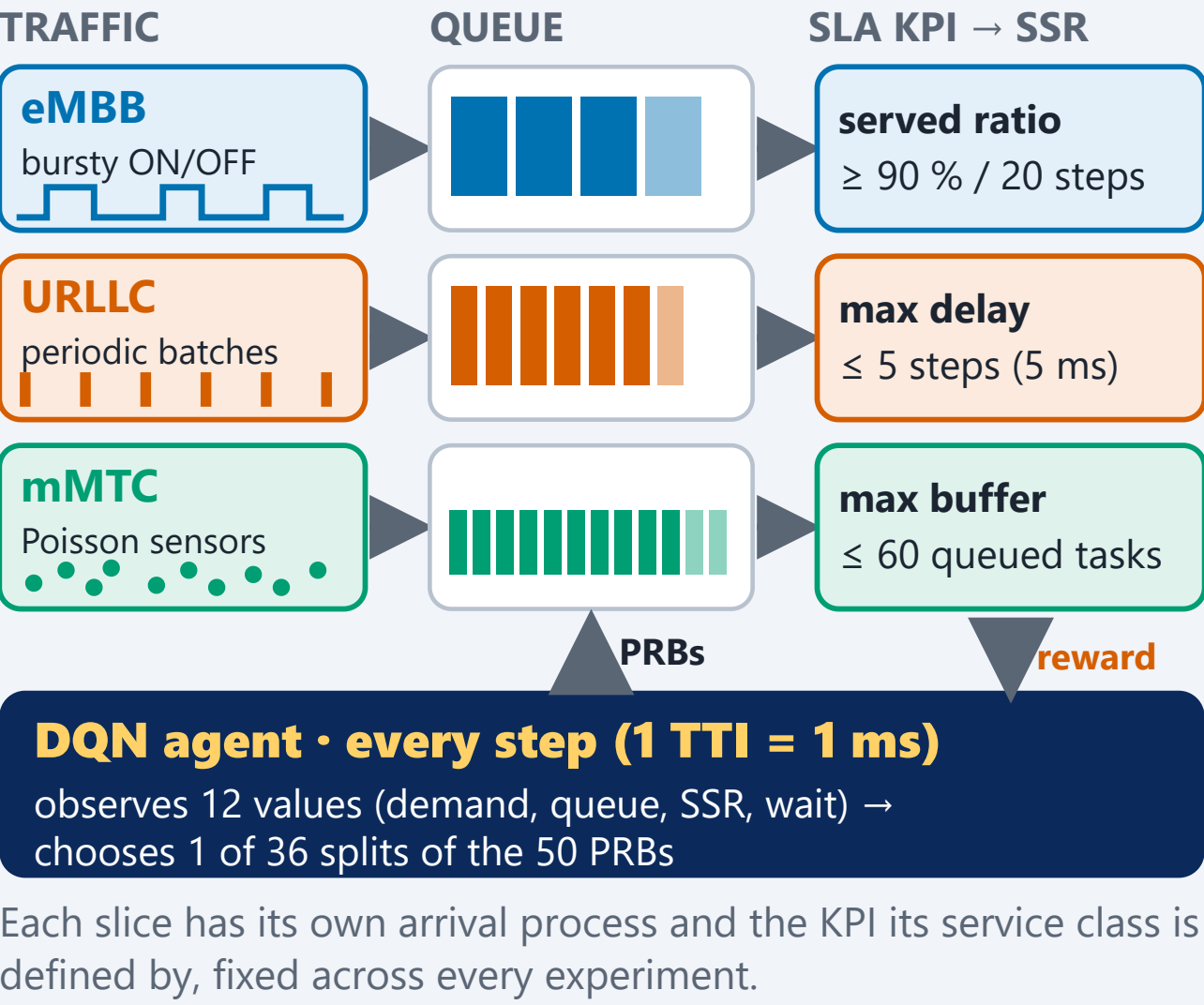
5G shares one radio cell between services with incompatible needs: **eMBB** wants throughput, **URLLC** wants low delay, **mMTC** wants to absorb many tiny reports.

A scheduler must split a fixed pool of **50 Physical Resource Blocks (PRBs)** between these *slices* every millisecond. Reinforcement learning (RL) can learn this, but a learned policy is tied to the traffic it saw in training.

Literature gap: everyone assumes a penalty for traffic change, nobody puts a number on it. We measure it: we built a simulator, trained Deep Q-Network (DQN) agents and compared them with heuristics.

- **Research question:** how large is the penalty, how does it compare with training noise, and can allocation-distribution distance predict it?

2 The RAN slicing simulator



3 Four load-matched environments

All four offer **48 of 50 PRB/step** (96 % utilisation); **only the mix differs**, so a measured gap can be blamed on traffic mix alone.

Environment	Offered load (PRB/step)		
Balanced	16	16	16
eMBB-int	24	12	12
URLLC-int	12	24	12
mMTC-int	12	12	24

- eMBB** **URLLC** **mMTC** (the chart colours used throughout).
- Arrival rates are **derived** from the load target, not tuned.
 - SLA thresholds and observation scaling are **global**: otherwise “SSR” would mean different things per environment.
 - The three intensive mixes sit at equal distance from balanced.

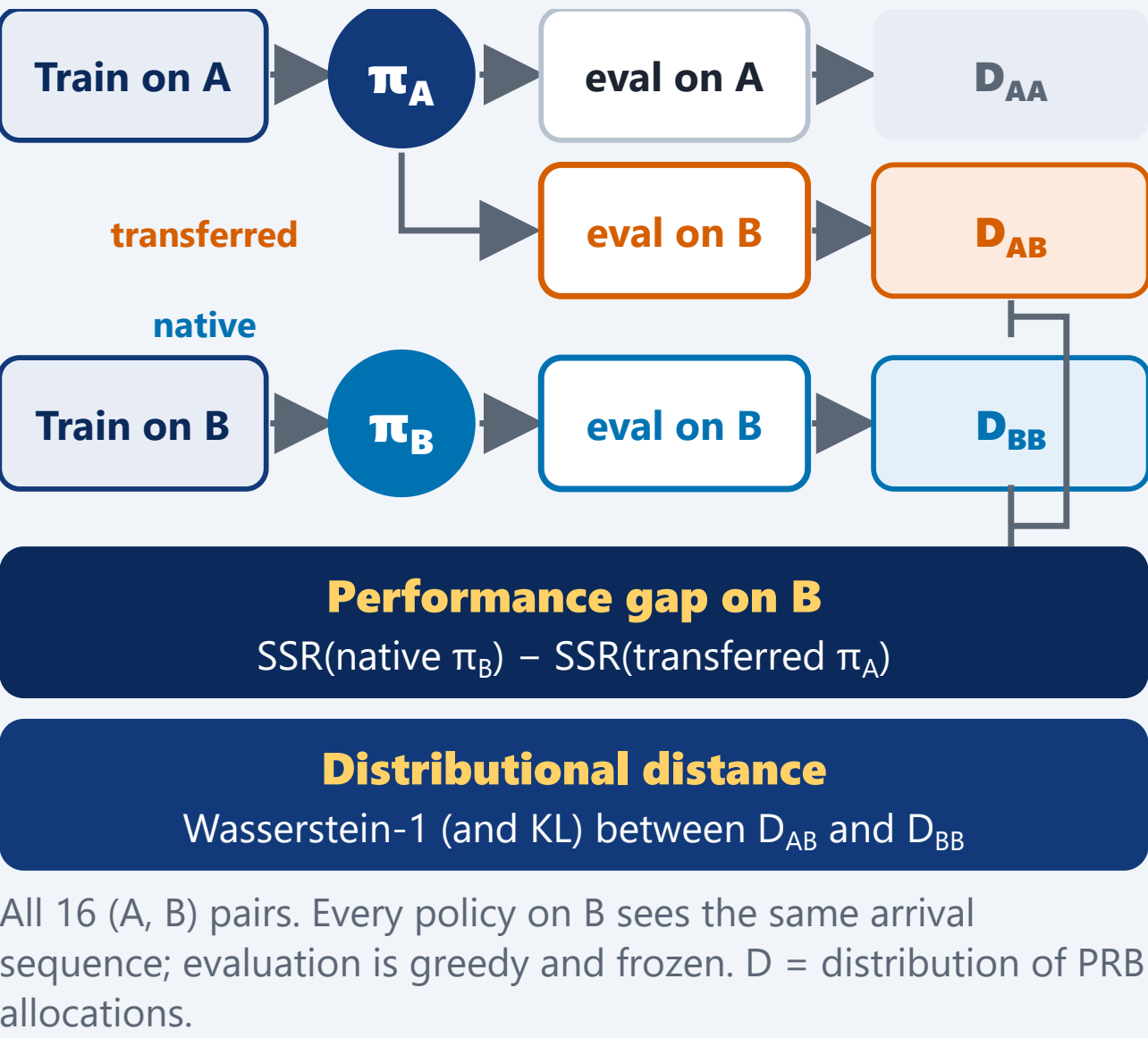
4 Reward: justified, not blind

$$R(t) = \sum_i SSR_i(t) - 0.1 \cdot \sum_i \hat{q}_i(t)$$

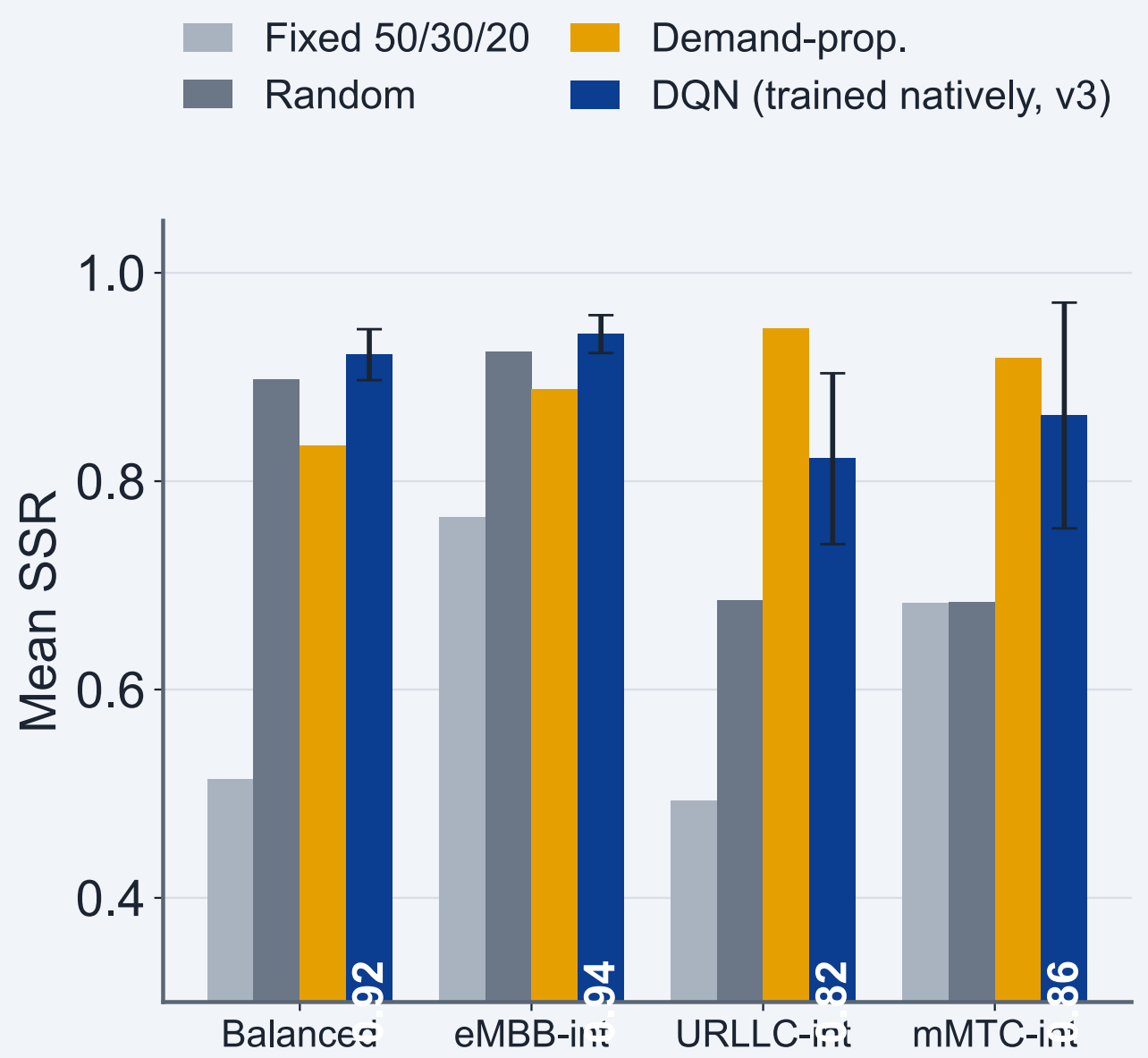
- **SSR** = SLA satisfaction ratio (0–1) per slice, so the reward matches the evaluation metric.
- **Rejected:** weighted SE + SSR sums, which can “blindly sacrifice SLA” for efficiency (Li et al.).
- **SSR × SE** is not computable here: with no channel model, spectrum efficiency is constant and the product collapses to SSR.
- **Equal slice weights:** any other weighting would be an unjustified priority claim.
- **Backlog term ($\hat{q}$):** credit assignment only. mMTC’s long-timescale KPI gives almost no gradient otherwise. The weight 0.1 was not swept.

- **Agent:** DQN 12→64→64→36, $\gamma = 0.99$, 5 seeds × 600 episodes × 500 steps.

5 Transfer protocol



6 RL vs. heuristics



- Mean SSR. DQN = natively trained, 5 seeds (± 1 SD). Demand-prop. splits PRBs by current backlog.
- **DQN beats the fixed split everywhere, but not the best simple heuristic in URLLC-int or mMTC-int.**

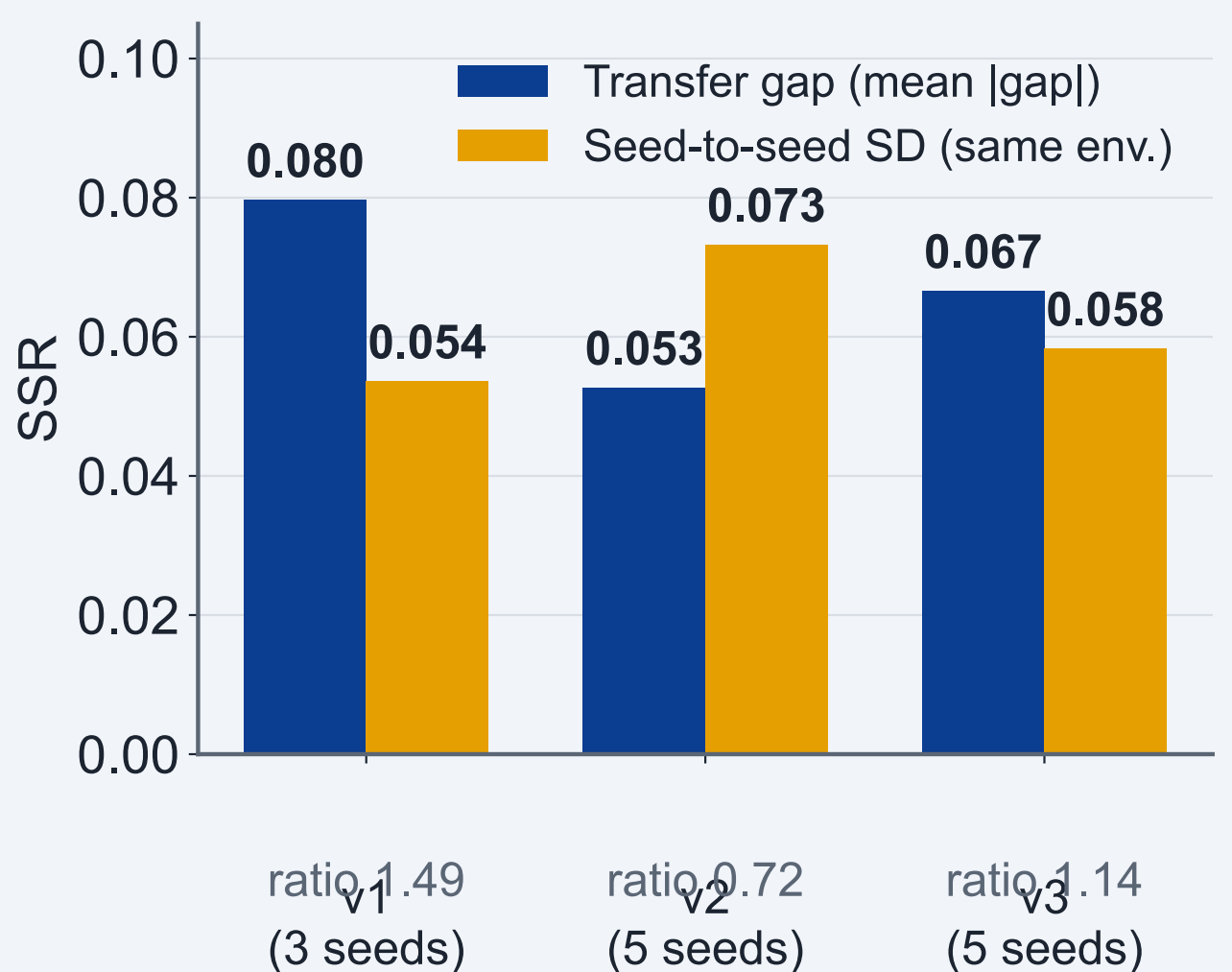
7 Transfer-gap matrix

		Evaluated on B			
		Balanced	eMBB-int	URLLC-int	mMTC-int
Trained on A	Balanced	native ref.	+0.016	-0.100*	-0.066
	eMBB-int	+0.057	native ref.	-0.021	+0.107
	URLLC-int	+0.061	+0.092	native ref.	+0.137*
	mMTC-int	-0.014	-0.023*	-0.104*	native ref.

Run v3 (600 episodes, 5 seeds). Red: native policy wins. Blue: transferred policy wins. * = $|\text{gap}| > 2$ standard errors.

- **Gaps reach +0.14 (URLLC → mMTC) but many are negative.**

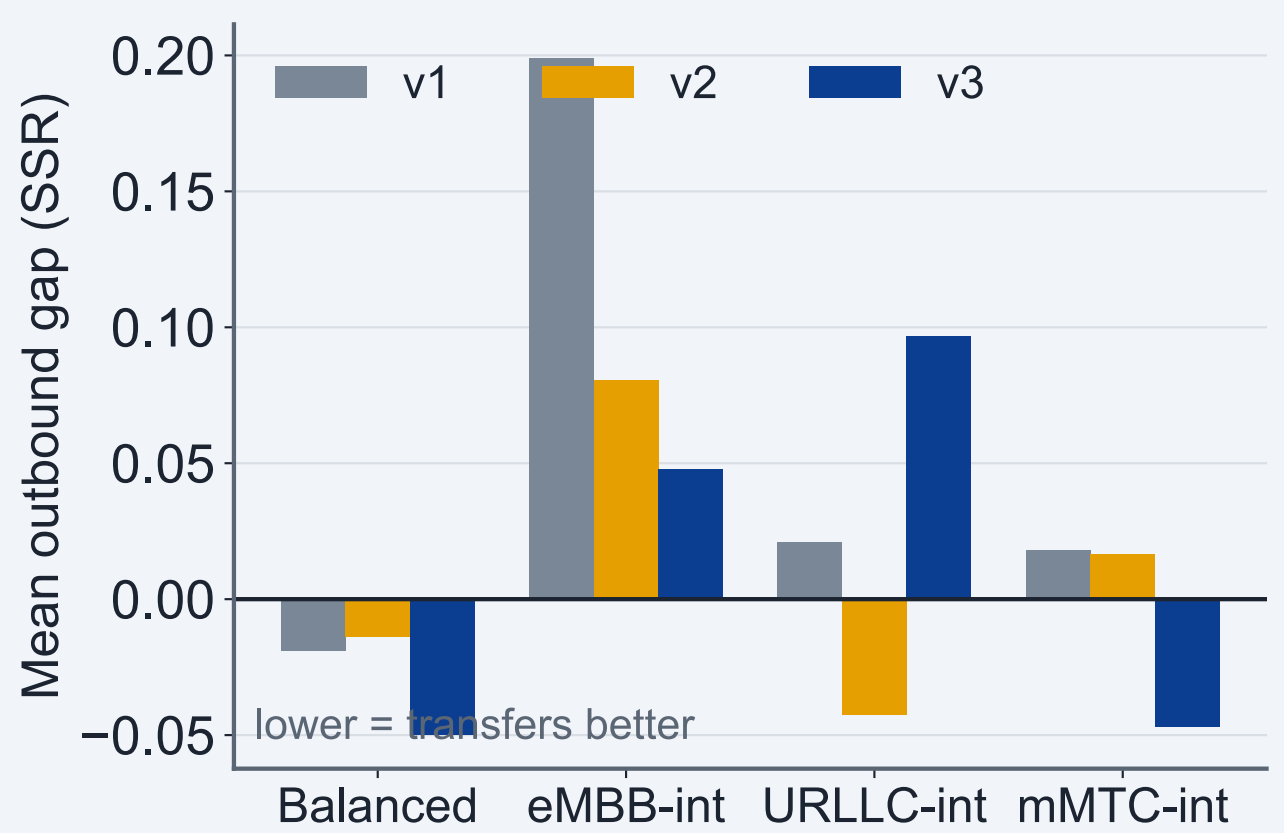
8 Gap vs. training noise



v1: 3 seeds, absolute-rate eMBB KPI. v2: 5 seeds, served-ratio KPI. v3: 600 episodes.

- **Retraining with a new seed moves SSR about as much as deploying in the wrong environment.**

9 Who transfers best?



Mean gap of a policy trained on the source environment and deployed on the other three.

- **Balanced traffic gives the most transferable policy in all three runs (rank 1 / 2 / 1).**